

M&A; 2026 Final Practice Study Guide

This guide is based on all materials in this repo, including:

- Practice finals and problem solutions
- Module notes (2 to 10)
- Mini-cases (DOCX)
- Modeling solution files (XLSX)
- Final review images

Use this as a fast revision sheet for likely MCQ and problem-style exam questions.

1) Quick Formula Sheet (Most Tested)

1 Enterprise Value

- $EV = \text{Equity Value} + \text{Debt} + \text{NCI} + \text{Preferred} - \text{Cash}$

1 Equity Value

- $\text{Equity Value} = EV - \text{Debt} - \text{NCI} - \text{Preferred} + \text{Cash}$

1 Multiples

- $EV/EBITDA = EV / EBITDA$
- $P/E = \text{Share Price} / \text{EPS}$

1 CAPM and WACC

- Cost of Equity: $Re = Rf + \beta_e * MRP$
- $WACC = W_e * Re + W_d * R_d * (1 - T_c)$

1 Unlever/Relever Beta (common structure)

- $\beta_u = \beta_e / [1 + (1 - T_c) * D/E]$
- $\beta_e = \beta_u * [1 + (1 - T_c) * D/E]$

1 Gordon Growth DCF

- $\text{Firm Value} = FCFF_1 / (WACC - g)$

1 Merger Gains and Offer Bounds

- $\text{Total Merger Gain} = \text{Combined Value} - \text{Standalone Acquirer Value} - \text{Standalone Target Value}$
- $\text{Max Offer Price (total)} = \text{Target Standalone Equity Value} + \text{Total Merger Gain}$
- $\text{Min Offer Price (total)} = \text{Target Standalone Equity Value}$

1 Acquisition Premium

- $\text{Premium (\$)} = \text{Offer Price} - \text{Target Standalone Equity Value}$
- $\text{Premium (\%)} = (\text{Offer Price} / \text{Target Standalone Equity Value}) - 1$

1 Goodwill

- $\text{Goodwill} = \text{Purchase Price} - \text{Fair Value of Net Identifiable Assets}$

1 LBO IRR

- Equity IRR comes from equity cash flows (initial negative equity check, later positive exit proceeds/dividends)

1 Mixed Consideration (Cash + Stock)

- Stock value paid per target share = Exchange Ratio * Acquirer New Share Issue Price
- Cash per target share = Total Offer per Target Share - Stock Value Paid per Target Share
- Cash proportion = Cash per target share / Total Offer per target share

2) Likely MCQ/Short Questions with Answers

Module 2: Multiples and Premiums

Q1. Which metric is equity-based (not firm-based)?

- A) EV/EBITDA
- B) EV/Sales
- C) P/E
- D) EV/EBIT

Answer: C

Simple explanation: P/E uses share price and earnings to equity holders.

Q2. If cash increases (all else equal), EV will usually:

- A) Increase
- B) Decrease
- C) Stay unchanged
- D) Become zero

Answer: B

Simple explanation: EV subtracts cash.

Q3. A higher acquisition premium usually means:

- A) Lower value transferred to target shareholders
- B) More value transferred to target shareholders
- C) No relation to value sharing
- D) Guaranteed acquirer gain

Answer: B

Simple explanation: premium is extra payment above target standalone equity value.

Q4. Best comparables should match mostly on:

- A) Industry, business model, growth, risk, size
- B) Headquarters city only
- C) Auditor only
- D) Dividend policy only

Answer: A

Simple explanation: valuation multiples are meaningful only with similar economics.

Module 3: DCF

Q5. In target valuation for M&A, start from:

- A) Target assets and cash flow claims
- B) Acquirer dividends
- C) Exchange ratio
- D) Voting rights only

Answer: A

Simple explanation: acquirer buys business assets and their cash flows.

Q6. In FCFF DCF, discount FCFF using:

- A) Cost of equity
- B) Cost of debt
- C) WACC
- D) Dividend yield

Answer: C

Simple explanation: FCFF belongs to all capital providers.

Q7. If long-run g rises while WACC fixed, DCF value:

- A) Falls
- B) Rises
- C) Unchanged
- D) Cannot be valued

Answer: B

Simple explanation: denominator $WACC - g$ becomes smaller.

Module 4: Other Valuation Issues

Q8. For private targets with no traded beta, a common approach is:

- A) Use acquirer beta directly
- B) Pure-play comparables and re-lever
- C) Use zero beta
- D) Use accounting ROE as beta

Answer: B

Simple explanation: unlever comparable betas then relever to target capital structure.

Q9. Size/illiquidity discounts are usually discussed in:

- A) Public mega-cap valuation only
- B) Private company valuation adjustments
- C) Accounting entries only

- D) PMI stage only

Answer: B

Simple explanation: private shares are less liquid and often require discount adjustments.

Module 5: Synergies

Q10. Synergies are best defined as:

- A) Cost of the adviser team
- B) Extra value from combining firms beyond standalone sum
- C) Target standalone value
- D) Debt tax shield only

Answer: B

Simple explanation: synergy = 1 + 1 > 2 value effect.

Q11. Typical synergy channels include:

- A) Cost savings, revenue uplift, financing benefit
- B) Tax filing only
- C) Board independence only
- D) Weather shocks only

Answer: A

Simple explanation: these are exactly modeled in module 5 spreadsheets.

Q12. If total synergy value is 22.38 and target standalone equity is 95.12, max total offer is:

- A) 72.74
- B) 95.12
- C) 117.50
- D) 22.38

Answer: C

Simple explanation: max offer = target standalone equity + total merger gain.

Module 6: Offer Price and Due Diligence

Q13. Minimum acceptable offer (seller side) is usually near:

- A) Target standalone equity value
- B) Acquirer standalone value
- C) Combined value with full synergies to acquirer
- D) Zero

Answer: A

Simple explanation: seller should not accept below standalone equity value.

Q14. Due diligence is primarily to:

- A) Increase fees only
- B) Validate assumptions/risks and reduce overpayment risk

- C) Replace legal documents
- D) Remove need for valuation

Answer: B

Simple explanation: DD checks quality of earnings, liabilities, synergies, and integration risk.

Module 7: Deal Terms and PMI

Q15. In stock acquisition, acquirer buys:

- A) Specific assets only
- B) Voting shares/ownership control
- C) Debt only
- D) Board minutes

Answer: B

Simple explanation: stock deals transfer ownership/control of the company.

Q16. Exchange ratio indicates:

- A) Cash interest rate
- B) New acquirer shares per target share
- C) Debt covenant headroom
- D) Tax rate

Answer: B

Simple explanation: used to compute stock consideration value.

Q17. PMI failure most often comes from:

- A) Perfect cultural fit
- B) Integration execution and culture mismatch
- C) Too much synergy
- D) Too little accounting detail only

Answer: B

Simple explanation: operational/cultural integration is a common deal value destroyer.

Module 8: M&A; Modeling and Tactics

Q18. Goodwill appears when:

- A) Purchase price < fair net assets
- B) Purchase price > fair net identifiable assets
- C) Cash exceeds debt
- D) EPS accretion is negative

Answer: B

Simple explanation: excess purchase price is recorded as goodwill.

Q19. Acquisition premium and goodwill are:

- A) Always equal

- B) Unrelated in all cases
- C) Related but not always equal
- D) Never both positive

Answer: C

Simple explanation: premium is market-reference; goodwill is accounting fair-value reference.

Q20. A common target defense tactic is:

- A) Poison pill
- B) CAPM
- C) Terminal value formula
- D) LBO recap model

Answer: A

Simple explanation: poison pills can deter hostile takeovers.

Module 9: LBO

Q21. LBOs are typically characterized by:

- A) Low debt use
- B) High debt financing and equity sponsor contribution
- C) No collateral
- D) No exit planning

Answer: B

Simple explanation: debt is a major funding source, often secured by target assets.

Q22. Higher exit EBITDA multiple (all else equal) generally:

- A) Lowers sponsor IRR
- B) Raises sponsor IRR
- C) No effect on IRR
- D) Eliminates debt

Answer: B

Simple explanation: higher exit valuation increases equity proceeds.

Module 10: Sell-offs and Revision

Q23. A sell-off is best described as:

- A) Buying a new division
- B) Selling a division/subsidiary to contract scope
- C) Debt refinancing
- D) Equity rights issue

Answer: B

Simple explanation: opposite direction from expansionary M&A.

Q24. Sell-offs may be value-enhancing when:

- A) Parent divests underperforming/non-core assets
- B) Parent has no strategic focus
- C) Synergies are guaranteed
- D) Debt cost is always zero

Answer: A

Simple explanation: sharper focus and better capital allocation can improve value.

3) High-Probability Problem Types (with solved templates)

Problem Type A: Mixed Payment (Cash + Stock)

Likely from your practice final/review image style.

Given:

- Acquirer new-share issue price = 20
- Offer per target share = 40
- Exchange ratio = 0.8

Step 1: Stock value paid per target share

- $0.8 * 20 = 16$

Step 2: Cash per target share

- $40 - 16 = 24$

Step 3: Cash proportion

- $24/40 = 60\%$

Final answer:

- Cash consideration proportion = 60%

Simple explanation:

- First find value of stock piece in the offer, then the remainder is cash.

Problem Type B: Synergy and Max Offer

Likely from module 5/6 and final problem part.

Given:

- Standalone firm value (target) = 104.64
- Target debt MV = 9.52
- Synergy (merger gain) = 22.38

Step 1: Target standalone equity

- $104.64 - 9.52 = 95.12$

Step 2: Maximum total offer

- $95.12 + 22.38 = 117.50$

Final answer:

- Maximum total offer price = 117.50

Simple explanation:

- Acquirer should not pay more than standalone equity plus total gains from merger.

Problem Type C: Constant-Growth FCFF DCF

Given:

- $FCFF_1 = 10$
- $WACC = 9\%$
- $g = 4\%$

Value:

- $V_{firm} = 10 / (0.09 - 0.04) = 200$

If debt = 50 and cash = 5:

- $Equity = 200 - 50 + 5 = 155$

Simple explanation:

- FCFF gives firm value first, then move to equity by subtracting net debt.

Problem Type D: EV from Market Inputs

Given:

- Shares = 20
- Share price = 5
- Debt = 50
- Cash = 5
- NCI = 0

Step:

- $Equity\ value = 20 * 5 = 100$
- $EV = 100 + 50 - 5 = 145$

Final answer:

- Enterprise value = 145

Simple explanation:

- EV is value of operating assets to all capital providers.

Problem Type E: Relevering Beta and WACC for Private Target

Given comparable average levered beta and D/E, first unlever:

- $\beta_u = \beta_e / [1 + (1 - T_c) * D/E]$

Then relever to target D/E:

- $\beta_{e,target} = \beta_u * [1 + (1 - T_c) * D/E_{target}]$

Then:

- $Re = R_f + \beta_{e,target} * MRP$

- $WACC = W_e \cdot R_e + W_d \cdot R_d \cdot (1 - T_c)$

Simple explanation:

- Remove financing effect from comparable betas, then add target financing effect back.

Problem Type F: LBO Equity IRR Skeleton

- 1 Entry: compute equity check = purchase price - debt financing.
- 2 During hold: debt paydown and any distributions.
- 3 Exit equity value = exit enterprise value - net debt at exit.
- 4 Compute IRR from equity cash flows.

Simple explanation:

- LBO IRR is a return on sponsor equity after leverage and deleveraging effects.

4) Fast Decision Rules for MCQ

- 1 If cash changes, check EV formula sign first.
- 2 If question says FCFF, discount by WACC (not R_e).
- 3 Offer range logic:
 - lower bound around target standalone equity
 - upper bound around target standalone equity + merger gains
- 1 Premium != goodwill in many cases.
- 2 In stock deals, exchange ratio gives shares, then multiply by acquirer share value.
- 3 If private target beta needed, use pure-play comparables.
- 4 LBO questions often test leverage effect on equity IRR.
- 5 Due diligence is risk-validation, not optional formality.

5) Last-Night Revision Checklist

- 1 Rework at least one question each from:
 - Multiples/EV bridge
 - DCF FCFF
 - Offer range and premium split
 - Synergy valuation
 - Exchange ratio mixed consideration
 - LBO IRR logic
- 1 Memorize these 8 formulas:
 - EV bridge
 - Equity bridge
 - CAPM
 - WACC
 - Gordon FCFF value

- Unlever beta
- Relever beta
- Max offer = target standalone equity + merger gains

1 In every problem, write units clearly:

- per share vs total value
- million vs dollars
- firm value vs equity value

1 In every final answer, check reasonableness:

- offer should lie within negotiation range unless question states overpayment
- cash proportion must be between 0% and 100%
- if g is close to WACC, value becomes very sensitive

If you want, I can next generate a one-page "night-before memory sheet" with only formulas + 20 ultra-likely question-answer pairs for quick cramming.